

Benchmarking Explanation Methods on a Multi-Task Vision Transformer

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Why Explainability?

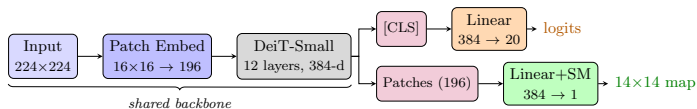
- Deep networks are powerful but opaque: we cannot easily see *why* a model makes a prediction.
- **Post-hoc explanation methods** produce attribution maps showing which input regions mattered most.
- Problem: many methods exist, but how do we know which explanation is “correct”?
- There is no universal ground truth for what a model “should” attend to.

Our idea: Use human gaze data (SALICON) as a quantitative behavioral reference for one dimension of explanation quality.

What Makes This Setup Unique?

- We train a multi-task ViT that *both* classifies images *and* predicts human saliency.
- This gives us two kinds of “where the model looks”:
 - The saliency head: what the model *explicitly predicts* humans look at.
 - XAI methods: what the model *implicitly relies on* for classification.
- SALICON fixation maps tell us where humans *actually* look.
- We can measure both **faithfulness** (does masking highlighted regions change the prediction?) and **human alignment** (do explanations match where people look?).

Multi-Task Architecture



- **DeiT-Small** (Data-efficient Image Transformer): a ViT trained with an improved recipe on ImageNet-1K without extra data. 12 layers, 384-d, 22M params.
- Classification head on CLS token $\rightarrow$ 20 COCO category logits.
- Saliency head on patch tokens $\rightarrow$ softmax $\rightarrow 14 \times 14$ distribution.
- Trained with KL divergence for saliency (CC = 0.899, mAP = 0.604).

SALICON + COCO

- **SALICON**: 10,000 train / 5,000 val COCO images with dense fixation maps.
- Fixation maps collected via mouse-contingent gaze tracking.
- **COCO 2014**: bounding-box annotations → top 20 category labels.
- Ground-truth saliency normalized to 14×14 distributions matching the ViT patch grid.

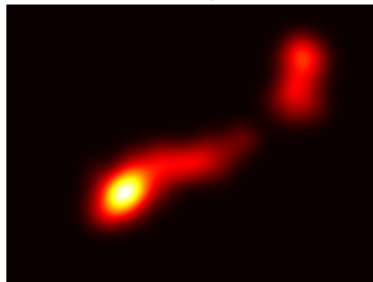
Key advantage: SALICON provides *behavioral* ground truth (thousands of observers), not subjective ratings.

What the Data Looks Like

Original Image



Human Gaze Map (SALICON)



Left: original image. Right: SALICON fixation map showing where thousands of observers looked. This behavioral data serves as our human-alignment reference for evaluating explanations.

Gradient-Based Methods

All explanations computed w.r.t. the model's most confident predicted class.

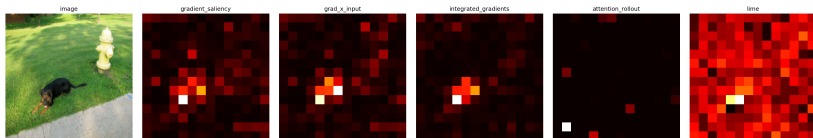
- **Gradient saliency** (Simonyan et al., 2014): $|\partial\ell/\partial x|$, max-pooled across channels. Fast but noisy.
- **Gradient $\times$ input** (Shrikumar et al., 2017): gradient $\odot$ input. Suppresses noise in dark regions.
- **Integrated gradients** (Sundararajan et al., 2017): accumulates gradients from a zero baseline (50 steps). Attributions are guaranteed to sum to the full output difference.

Architecture and Perturbation Methods

- **Attention rollout** (Abnar & Zuidema, 2020): multiplies attention matrices across 12 layers with residual mixing ($0.5A + 0.5I$). No gradients needed.
- **LIME** (Ribeiro et al., 2016): local linear surrogate on 300 perturbations over a 14×14 patch grid. Model-agnostic.

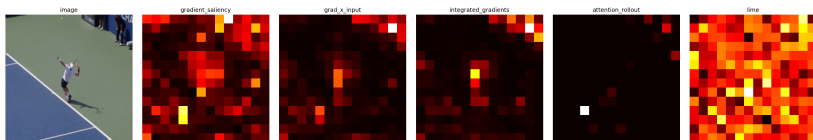
Three families: **gradient** (how sensitive is the output to each pixel?), **architecture** (where does the model route information?), **perturbation** (what happens when we hide parts of the input?).

Visual Comparison: Dog + Fire Hydrant



Gradient methods highlight the dog. LIME highlights a broader region. Attention rollout collapses to a single bright patch.

Visual Comparison: Tennis Player

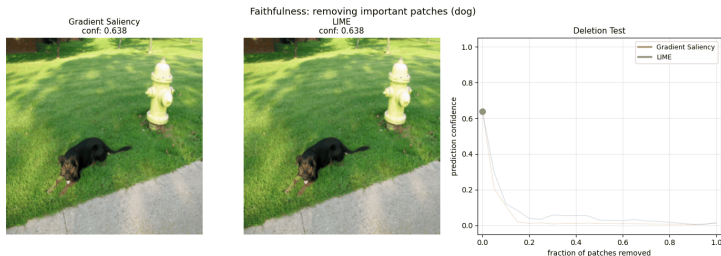


Gradient methods agree on the player's location. LIME assigns importance to the court surface, suggesting the model uses background context.

Three Evaluation Axes

- ❶ **Faithfulness** (does the explanation reflect what the model actually uses?)
 - **Deletion test:** mask patches in importance order, measure confidence drop. Lower AUC = better.
 - **Insertion test:** reveal patches in importance order from blank. Higher AUC = better.
- ❷ **Stability** (are explanations consistent under small perturbations?)
 - Cosine similarity under Gaussian noise, horizontal flip, brightness shift.
- ❸ **Human alignment** (does the attribution match where people look?)
 - CC and SIM against SALICON fixation maps.

Deletion Test: Live Demo



Patches are masked in order of attribution importance (Gradient Saliency vs LIME). The confidence curves show how quickly each method identifies decision-critical regions.

Main Results Table

Method	Del AUC ↓	Ins AUC ↑	CC ↑	SIM ↑
Gradient saliency	0.468	0.766	0.161	0.410
Grad × input	0.521	0.768	0.171	0.405
Integrated gradients	0.479	0.778	0.248	0.434
Attention rollout	0.544	0.745	−0.003	0.123
LIME	0.409	0.881	0.107	0.416

- LIME: best faithfulness, worst human alignment among gradient methods.
- Integrated gradients: best human alignment and stability.
- Attention rollout: collapses, near-zero alignment.

The Central Finding: Faithfulness $\neq$ Human Alignment

- LIME is the most **faithful** because it directly models the classifier's local decision boundary.
- But what the classifier relies on is *not* what humans look at.
- The model may use texture, context, or background cues invisible to human inspection.
- **Integrated gradients** strikes the best balance: moderate faithfulness, highest human alignment, highest stability.

Implication: choosing an XAI method depends on the goal (model debugging vs. human communication).

Attention Rollout Collapse

- Attention rollout was designed for shallow transformers.
- In DeiT-Small (12 layers), multiplying attention matrices with residual mixing ($0.5A + 0.5I$) causes exponential compounding of CLS-token asymmetry.
- Result: rollout concentrates nearly all weight on 1–2 patches regardless of image content.
- This is consistent with numerical degeneration in deep attention rollout, not a bug in our code.
- $CC \approx -0.003$: no systematic relationship with human gaze.

Stability Results

Method	Noise 0.02	Noise 0.05	H-flip	Brightness
Gradient saliency	0.963	0.918	0.768	0.988
Integrated gradients	0.990	0.969	0.820	0.999

- Integrated gradients is more stable across all perturbation types.
- Path integration acts as a smoothing operation, reducing sensitivity to small input changes.
- Both methods are highly stable under brightness shifts (DeiT features are intensity-invariant).

Key Takeaways

- ① **Faithfulness and human alignment are distinct properties**, even inversely related in some cases.
- ② **LIME** is the most faithful (best deletion/insertion AUC) but does not match human gaze patterns.
- ③ **Integrated gradients** is the best practical default: moderate faithfulness, best alignment, best stability, axiomatic guarantees.
- ④ **Attention rollout** failed in our DeiT-Small setup, consistent with numerical degeneration of deep matrix products.
- ⑤ **Method choice depends on use case**: model debugging → faithfulness (LIME); human communication → alignment (IG).

Limitations and Future Work

- Single model (DeiT-Small) and dataset (SALICON/COCO); results may vary across architectures.
- 100-sample evaluation provides stable means but wide per-image variability.
- Attribution maps and fixation maps both downsampled to 14×14 , which may obscure fine-grained differences.
- Future: test on CNNs and larger ViTs, higher-resolution attributions, per-category analysis, additional XAI methods (SmoothGrad, GradCAM).

Questions?

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github.com/nickvpn/vit-ml-xai-project